

Contrasting Incorrect and “Don’t Know” Responses to Personal Finance Questions in Relation to Financial Behavior and Financial Education

Introduction

When faced with a financial question a person may answer correctly, answer incorrectly, or state that they don’t know the answer. Recent research has focused on “don’t know” responses, investigating whether some people who say they don’t know, in fact do know the answer, and the potential improvements in behavior if such people were to answer correctly instead (Bucher-Koenen et al. 2021).

This study will address a question that has received less attention: the differences in behavior that come from answering “don’t know” as opposed to answering incorrectly. It is plausible to think that answering incorrectly instead of understanding one doesn’t know indicates over-confidence and could lead to negative financial behavior.

We also examine the role of financial education, as this has often been proposed as a way to improve financial knowledge (Kaiser et al. 2022). Previous research has typically focused on whether taking a financial education course increases the number of correct responses, but has not looked at whether it decreases the number of incorrect responses (the former does not entail the latter when there is the option to answer “don’t know”).

Our analysis suggests that answering incorrectly rather than answering “don’t know” is associated with an increase in problematic financial behavior. Therefore, we should be cautious in suggesting people trust they know the answer to financial questions, as if it pushes them to answer incorrectly it could have harmful consequences.

We also find that taking a course in financial education is associated with an increase in the number of incorrect responses a person provides. This suggests there are some previously

under-appreciated risks to financial education as it is currently being provided. Identifying method in financial education that reduce participants' tendency to mistakenly think they know the answer to a financial questions is therefore a pressing task for improving consumer well-being.

Background

When assessing financial literacy, a person is often presented with knowledge questions to which three outcomes are possible: they provide a correct response, they provide an incorrect response or they state they don't know the answer. Traditionally, research has treated "don't know" and incorrect responses interchangeably (Bucher-Koenen et al. 2021).

However, recent research has started to examine the unique features of "don't know" responses. Wilmarth, Kim, and Pak (2023) find that responding "don't know" is associated with personality traits plausibly connected to confidence – for example, those with positive affect are less likely to select don't know, while those with negative affect are more so (Tekinsoy, Tunç, and Ülgen 2024). Further, Bucher-Koenen et al. (2021) find that those selecting "don't know" are less likely to engage in investment behavior while Wilmarth, Kim, and Pak (2023) find that they are less likely to have an emergency fund or plan for retirement.

Bucher-Koenen et al. (2021) used a dataset in which respondents were asked the same set of financial literacy questions on two separate occasions, the first time with a "don't know" option, the second time without. They found that there was a significant amount of knowledge among those answering "don't know". That is, people who at first answered "don't know", answered correctly more frequently when required to select an answer than could be explained by lucky guessing (see also, N. Kim and Mountain 2019; Pearson et al. 2024).

The researchers take from this an imperative to reduce people's tendency to think they don't know the answer to financial questions: "From this point of view, it seems crucial to support individuals not only in acquiring financial knowledge but also in instilling confidence in their knowledge." (Bucher-Koenen et al. 2021, 28)

This aligns with research that has found some negative behaviors associated with under-confidence, most notably around limited engagement with financial tools. Angrisani and Casanova (2021) finds that such individuals fare worse with regard to retirement preparations. Ahmad (2020) finds that under-confident investors in developing countries experience worse financial outcomes. Huang et al. (2023) find that under-confidence negatively affects market participation, while Parker et al. (2012) find that low confidence is associated with worse retirement planning and fee minimization for investors.

When evaluating this conclusion, however, it's important to note that some of those selecting "don't know" really do not know the answer, and if they are persuaded to provide a response, may make an error. Limited research to date has investigated the difference in financial behavior associated with "don't know" and incorrect responses. Given that answering incorrectly when one has the option to say one doesn't know is plausibly related to over-confidence, it is important to establish whether this kind of response is associated with the same behavioral patterns as other measures of over-confidence.

A significant body of research has found that over-confident individuals engage in a range of problematic financial behaviors: for example, fall into mortgage delinquency (K. T. Kim, Lee, and Hanna 2020), trade investments more frequently (Abreu and Mendes 2012; Barber and Odean 2000), invest in cryptocurrencies more frequently (K. T. Kim, Hanna, and Lee 2022), use costly alternative financial services and credit cards (Mokhtari and Chawla 2023;

Robb et al. 2015) under diversify their investment portfolios (Goetzmann and Kumar 2008), and are less likely to seek out financial advice (Porto and Xiao 2016).

An additional factor to consider in relation to this is financial education, as this has often been proposed as a way to improve financial knowledge (Kaiser et al. 2022). It is plausible to think, following Bucher-Koenen et al. (2021), that this includes improved self-awareness of one's financial knowledge. In general, education has been thought to be a promising way to improve self-knowledge (Roeser and Peck 2009)

Understanding don't know responses is thus a crucial research question, especially as the frequency with which people provide such a response is increasing over time (K. T. Kim, Lee, and Hanna (2022)).

Purpose

The purpose of this study will first be to explore the significance of answering a financial question incorrectly in contrast to answering "don't know". Second, to investigate how taking a financial education course may be associated with the responses a person provides. This complements and fills in the gaps left by existing research that focuses on the contrast between don't know and correct responses.

We will look both at the behaviors associated with answering don't know compared to answering incorrectly, as well the relationship between taking a financial education course and what kind of responses a person makes to financial knowledge questions. Improving our understanding of these areas will allow us to better leverage financial knowledge and financial education to improve consumer well being.

Research Questions

We aim to address the following two research questions.

RQ1: Is answering don't know instead of answering incorrectly associated with an increase in beneficial financial behaviors, an increase in harmful financial behaviors, or an increase in risky investment behavior?

RQ2: Is taking a personal finance course associated with an increase in don't know responses relative to incorrect responses?

Methodology

Our study uses data from the 2024 National Financial Capability Survey (NFCS). This dataset is representative of the US adult population, with a sample size of 25,539. Respondents answer a large range of questions on their financial attitudes and behaviors. In addition, they are asked six multiple choice questions to test their financial knowledge on a range of topics. Respondents are given the option to choose an answer or select "don't know".

Variables and Analyses

When answering RQ1, our independent variables were based on the responses to the six financial knowledge questions. We have a score for knowledge, uncertainty, and error:

1. **Correct:** Number of correct responses to financial knowledge questions
2. **Don't Know:** Number of "don't know" responses to financial knowledge questions
3. **Error:** Number of incorrect responses to financial knowledge questions

Each of these variables can take an integer value between 0 and 6. Note that the sum of these values must always be six, so the three variables have two degrees of freedom. Therefore,

in a regression model, only two of the three variables can be included as independent variables, with the other one acting as a reference.

Therefore, we take **Don't Know** as the main independent variable, with **Error** as the reference class. We include **Correct** as an additional variable, controlling for the number of correct responses.

For RQ1, we have financial behavior as the outcome of interest. Looking through the survey questions, we found that the financial outcomes relevant to our study were naturally divided into positive and negative behaviors. We identified four key variables in each category to use in our model. (These are similar to those used in Marley-Payne, Dituri, and Davidson 2022).

The *positive variables* are: (i) having created an emergency fund; (ii) having opened a savings account; (iii) having opened an investment account; (iv) having made a plan for retirement (for current retirees, having made a plan for retirement before they retired).

The *negative variables* are: (i) occasionally overdrawing a checking account; (ii) being charged interest on a credit card balance in the past year; (iii) struggling to pay bills each month; (iv) using alternative financial services, such as a payday loan, in the past five years.

We take total *positive behaviors* and total *negative behaviors* as outcome variables (with values between 0 and 4). In addition, we identified a question asking if the respondent had bought *cryptocurrency* as an indication of risky investment behavior, which we take as a final outcome variable.

We include, as controls, key demographic variables including gender, race, age, and income, along with all variables used in the survey weighting, so we don't have to weight the regression analysis – reducing the standard errors in our results.

We created regression models analyzing each of the outcome variables in turn. For total *positive* and *negative* behaviors, we used OLS linear regression. For *cryptocurrency* use, we used logistic regression, as it is a binary outcome.

Turning to RQ2, we created a binary **financial education** variable taking value one if the respondent took a course in financial education, either in high school, college or at work, and zero otherwise. We constructed OLS linear regression models using this as our independent variable and *Correct*, *Don't Know* and *Incorrect* as outcomes. In each case, we used the full set of controls.

Results

Table 1 shows the results of the regression models addressing RQ1. We see that an increase in DK responses, relative to errors, significantly decreases the the predicted number of positive behaviors. However, it also significantly decreases the predicted number of negative behaviors and significantly decreases the probability of owning cryptocurrency.

On the other hand, an increase in correct responses, relative to errors, significantly increases the the predicted number of positive behaviors. In addition, it significantly decreases the predicted number of negative behaviors and significantly decreases the probability of owning cryptocurrency.

Table 2 displays regression results taking financial education as the independent variable. We see that taking financial education is associated with a decreased number of “don't know” responses and with an increased number of both correct and incorrect responses.

Discussion

These results present nuanced findings regarding the contrast between “don't know” and “incorrect” responses. They show that answering don't know is associated with a decrease in

positive behavior but also a decrease in negative behavior, compared to answering incorrectly. In other words, thinking you don't know the answer rather than mistakenly believing you know it comes with both benefits and drawbacks.

The result regarding crypto-currency is of particular significance. Prior research has shown that answering don't know is associated with increased investment behavior. Though investing in a diversified portfolio including stocks and other volatile financial instruments is a crucial part of building wealth over the long term, uninformed investing in overly risky products, such as some crypto-currency products, can cause significant financial harm. Therefore, the lower frequency of investment behavior associated with "don't know" responses is a double edged sword, as it involves avoiding potential pitfalls as well as missing out on potential benefits.

The results regarding financial education provide some cause for concern. Though taking a financial education course is associated with an increase in correct responses, which has beneficial associations across all types of behavior, it is also associated with an increase in incorrect responses, which comes with drawbacks. Understanding why this is so is a crucial topic for further research to ensure there are not areas in which financial education is working counter to its aims.

Limitations

The dataset used in this paper is observational and cross-sectional so we are unable to draw any causal inferences.

Implications for Practitioners

When practitioners are assessing the knowledge of clients, they should carefully distinguish instances in which the client claims they don't know the answer and when they

answer incorrectly, as these have divergent behavioral consequences. Where clients answer don't know, one should be on the look out for under-confidence while when they answer incorrectly, one should be on the look out for over-confidence.

In providing instruction, it is important not just to increase knowledge of financial matters but to increase self-knowledge. Strategies should be explored that allow consumers to assess whether they really know something when they think they don't know, and whether their belief they know the answer is faulty and they would be better assuming they don't know the answer.

While previous research has emphasized encouraging consumers to have faith in their knowledge rather than thinking that they don't know the answer, this study suggest the matter is more nuanced. Knowing that you don't know is not always a bad thing.

Tables

Table 1

	Positive Behaviors	Negative Behaviors	Owens Crypto
DK Responses	-0.12 *** (0.01)	-0.10 *** (0.01)	-0.34 *** (0.02)
Correct Responses	0.05 *** (0.01)	-0.13 *** (0.01)	-0.14 *** (0.02)
Female	-0.10 *** (0.02)	0.04 * (0.01)	-0.92 *** (0.05)
Black	0.10 *** (0.02)	0.17 *** (0.02)	0.27 *** (0.07)
Hispanic/Latino	0.01 (0.03)	0.08 ** (0.03)	0.18 ** (0.07)
Married	0.14 *** (0.02)	-0.08 *** (0.02)	-0.26 *** (0.05)
Low Income	-1.41 *** (0.03)	0.37 *** (0.03)	-1.01 *** (0.10)
N	22118	22597	23941
R2	0.36	0.19	
AIC	65420.74	65790.34	14063.92
BIC	65692.88	66063.21	14330.67
Pseudo R2			0.18

*** p < 0.001; ** p < 0.01; * p < 0.05.

Table 2

	DK Responses	Correct Responses	Incorrect Responses
Fin Ed	-0.59 *** (0.03)	0.39 *** (0.02)	0.20 *** (0.02)
Female	0.71 *** (0.02)	-0.61 *** (0.02)	-0.10 *** (0.01)
Black	0.12 *** (0.03)	-0.50 *** (0.03)	0.38 *** (0.02)
Hispanic/Latino	0.09 * (0.04)	-0.25 *** (0.03)	0.16 *** (0.03)
Married	-0.01 (0.02)	0.04 (0.02)	-0.02 (0.02)
Low Income	0.87 *** (0.05)	-1.01 *** (0.04)	0.14 *** (0.03)
N	25539	25539	25539
R2	0.18	0.28	0.06

*** p < 0.001; ** p < 0.01; * p < 0.05.

References

- Abreu, Margarida, and Victor Mendes. 2012. "Information, Overconfidence and Trading: Do the Sources of Information Matter?" *Journal of Economic Psychology* 33 (4): 868–81.
- Ahmad, Maqsood. 2020. "Does Underconfidence Matter in Short-Term and Long-Term Investment Decisions? Evidence from an Emerging Market." *Management Decision* 59 (3): 692–709.
- Angrisani, Marco, and Maria Casanova. 2021. "What You Think You Know Can Hurt You: Under/over Confidence in Financial Knowledge and Preparedness for Retirement." *Journal of Pension Economics and Finance* 20.
- Barber, Brad M., and Terrance Odean. 2000. "Trading Is Hazardous to Your Wealth: The Common Stock Investment Performance of Individual Investors." Rochester, NY: Social Science Research Network.
- Bucher-Koenen, Tabea, Rob J. Alessie, Annamaria Lusardi, and Maarten van Rooij. 2021. "Fearless Woman: Financial Literacy and Stock Market Participation." National Bureau of Economic Research.
- Goetzmann, William N., and Alok Kumar. 2008. "Equity Portfolio Diversification." *Review of Finance* 12 (3): 433–63. <https://academic.oup.com/rof/article-abstract/12/3/433/1598033>.
- Huang, Huichun, Junli Yuan, Guanghua Lin, and Jing Chi. 2023. "Underestimation of Financial Literacy and Financial Market Participation." *Journal of the Asia Pacific Economy* 28 (1): 75–100.
- Kaiser, Tim, Annamaria Lusardi, Lukas Menkhoff, and Carly Urban. 2022. "Financial Education Affects Financial Knowledge and Downstream Behaviors." *Journal of Financial Economics* 145 (2): 255–72.
- Kim, Kyoung Tae, Sherman D. Hanna, and Sunwoo T. Lee. 2022. "Investment Literacy, Overconfidence and Cryptocurrency Investment." Rochester, NY: Social Science Research Network.
- Kim, Kyoung Tae, Jonghee Lee, and Sherman D. Hanna. 2020. "The Effects of Financial Literacy Overconfidence on the Mortgage Delinquency of US Households." *Journal of Consumer Affairs* 54 (2): 517–40.
- Kim, Kyoung Tae, Sunwoo Lee, and Sherman D. Hanna. 2022. "Has Financial Knowledge Increased in the United States?" *Journal of Financial Counseling and Planning*.
- Kim, Namhoon, and Travis P. Mountain. 2019. "Financial Knowledge and 'Don't Know' Response." *Journal of Consumer Affairs* 53.
- Marley-Payne, Jack, Philip Dituri, and Andrew Davidson. 2022. "Financial Education, Mathematical Confidence, and Financial Behavior." *Journal of Financial Counseling and Planning* 33 (2): 194–204.
- Mokhtari, Mitch, and Isha Chawla. 2023. "Financial Literacy Overconfidence, Mobile Financial Service Use, and High-Cost Borrowing." *Journal of Financial Counseling and Planning* 34 (2): 184–203.
- Parker, Andrew M., Wändi Bruine De Bruin, Joanne Yoong, and Robert Willis. 2012. "Inappropriate Confidence and Retirement Planning: Four Studies with a National Sample." *Journal of Behavioral Decision Making* 25.
- Pearson, Blain, Thomas Korankye, Yi Liu, and Di Qing. 2024. "Knowing the 'Don't Knows' to Financial Literacy Questions in the US National Financial Capability Study." *Journal of Financial Counseling and Planning* 35 (2): 292–304.
- Porto, Nilton, and Jing Jian Xiao. 2016. "Financial Literacy Overconfidence and Financial Advice Seeking." *Journal of Financial Service Professionals* 70 (4).
- Robb, Cliff A., Patryk Babiarz, Ann Woodyard, and Martin C. Seay. 2015. "Bounded Rationality and Use of Alternative Financial Services." *Journal of Consumer Affairs* 49 (2): 407–35.
- Roeser, Robert W., and Stephen C. Peck. 2009. "An Education in Awareness: Self, Motivation, and Self-Regulated Learning in Contemplative Perspective." *Educational Psychologist* 44 (2): 119–36.
- Tekinsoy, Pelin Narin, Pervin Tunç, and Margörit Rita Krespi Ülgen. 2024. "Personality Traits, Attachment Styles, and Financial Attitudes." *Journal of Financial Counseling and Planning*, October.
- Wilmarth, Melissa J., Kyoung Tae Kim, and Tae-Young Pak. 2023. "What Do We Really Know about 'Don't Know'? Re-Assessing the Measurement of Financial Knowledge." *Journal of Consumer Affairs* 57 (4): 1623–49.